



**Killara**  
HIGH SCHOOL

STUDENT NUMBER

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# MATHEMATICS

## Extension 1

ASSESSMENT TASK NO: 3

Date: 10<sup>th</sup> June 2011

**TIME ALLOWED:** *Reading time: 5 minutes*  
*Working time: 60 minutes*

**INSTRUCTIONS:** This paper consists of 3 questions  
ANSWER all 3 questions  
Start each question in a new booklet  
Marks may not be given for untidy and poorly arranged work.  
All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

|     |                                                                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| HE3 | uses a variety of strategies to investigate mathematical models of situations involving simple harmonic motion, or exponential growth and decay |
| HE5 | applies the chain rule to problems including those involving velocity and acceleration as functions of displacement                             |
| HE4 | uses the relationship between functions, inverse functions and their derivatives                                                                |

|                                                                                           | MARKS |
|-------------------------------------------------------------------------------------------|-------|
| QUESTION 1 Applications of Calculus to Physical world                                     | /17   |
| QUESTION 2 Applications of Calculus to Physical world /Inverse<br>Trigonometric Functions | /15   |
| QUESTION 3 Inverse Trigonometric Functions                                                | /14   |
|                                                                                           | /46   |

*This assessment task constitutes 20% of the Higher School Certificate Course Assessment.*

**Question 1** (17 marks) [BEGIN A NEW PAGE]

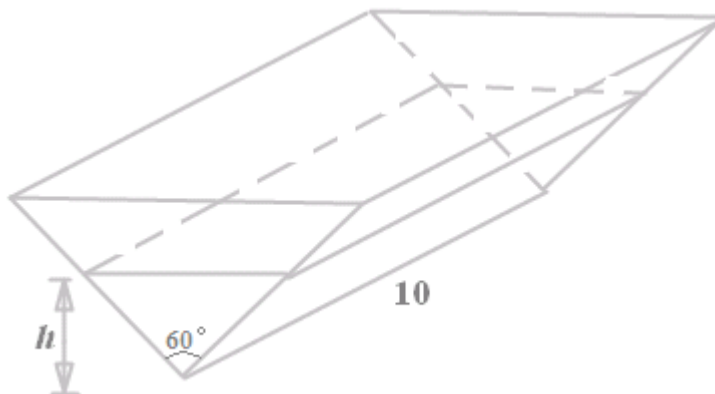
- a) The velocity of a particle  $P$  is given by the equation  $v = -2\sqrt{1-x^2}$ , where  $x$  is the displacement of  $P$  from the origin in metres after  $t$  seconds, with movement being in a straight line.

Initially the particle is at the origin.

- (i) Find an expression for the displacement of the particle in terms of  $t$ . 2

- (iii) Determine the velocity of the particle at time  $\frac{\pi}{6}$  seconds. 2

- b) A large irrigation trough in the shape of a triangular prism with base angle  $60^\circ$  is filled with water at a constant rate of  $2\text{m}^3/\text{min}$ . (*All lengths are in metres*).

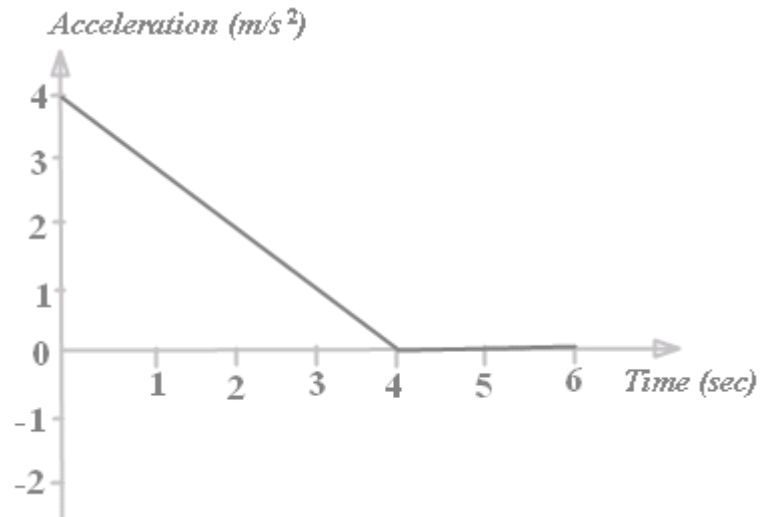


- (i) Show that the equation for the volume of water in the trough in terms of  $h$ , the depth of water in metres is given by  $V = \frac{10h^2}{\sqrt{3}}$ . 1

- (ii) Find the rate of change of depth of water when the depth is 2 metres. 2

**Question 1 continues...**

- c) The graph shows the acceleration of a particle during a 6 second interval.



Initially the particle is at rest at the origin.

- (i) Sketch the velocity – time graph of the particle for  $0 \leq t \leq 6$ . 3
- (ii) Calculate the distance travelled by the particle during this 6 seconds. 2

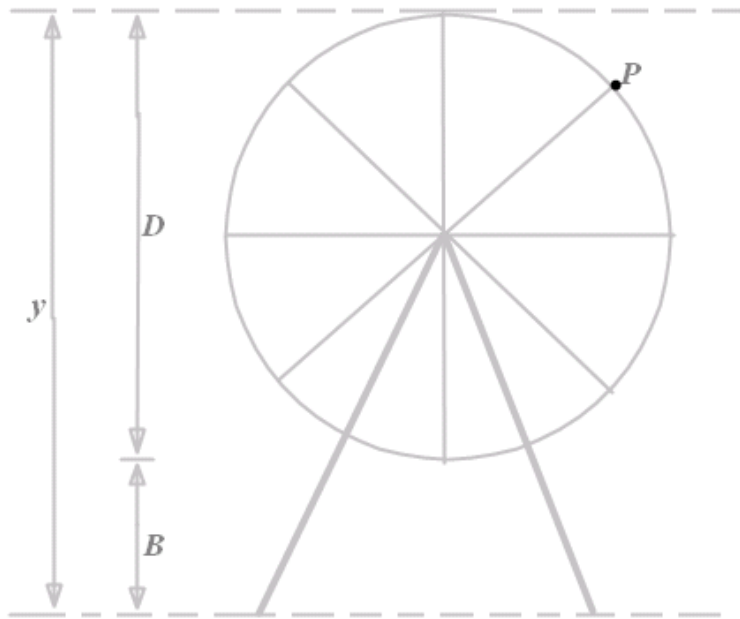
- d) A particle moves in a straight line with  $\ddot{x} = 3 - 2x$ . The velocity of the particle is  $v = 2 \text{ ms}^{-1}$  when the displacement  $x = 1 \text{ m}$ .

- (i) Find the velocity  $v$  of the particle in terms of  $x$ . 2
- (iii) Describe the motion of the particle. 3  
*(Draw sketches of the velocity and acceleration graphs to aid your explanation.)*

**End of Question 1**

**Question 2** (15 marks) [BEGIN A NEW PAGE]

- a) The diagram shows a giant Ferris wheel at a fair ground.  
 $B$  = the distance the bottom of the wheel is above the ground.  
 $D$  = the diameter of the wheel  
 $P$  is a point on the wheel.



The wheel rotates at a constant rate. The formula  $y = 25 + 18 \cos \left[ \frac{\pi}{10}(t + 9) \right]$  gives the height of the point  $P$  above the ground at time  $t$  seconds.

- |       |                                                                                                                |   |
|-------|----------------------------------------------------------------------------------------------------------------|---|
| (i)   | Prove that the height of the point $P$ is moving in simple harmonic motion and determine the centre of motion. | 2 |
| (ii)  | Calculate the diameter of the wheel and the height the bottom of the wheel is above the ground.                | 2 |
| (iii) | Determine the number of revolutions the wheel makes per minute.                                                | 2 |

**Question 2 continues....**

**Question 2 continued..**

b) Evaluate:

(i)  $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$  (in radians) **1**

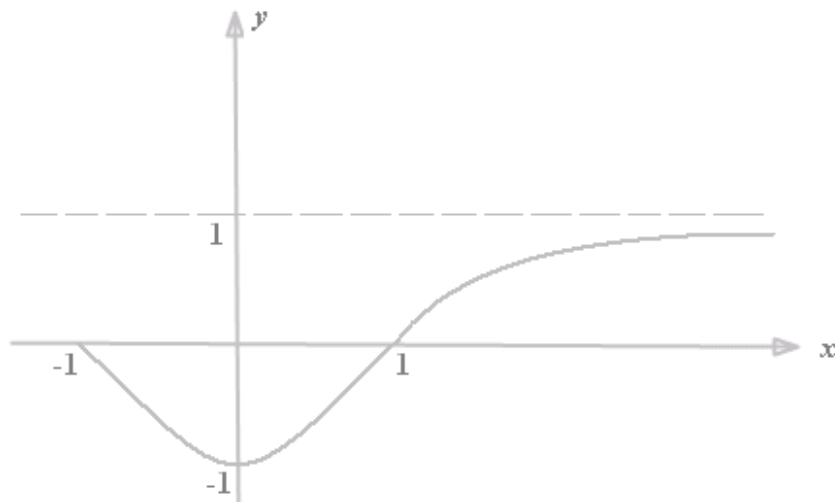
(ii)  $\tan\left(\cos^{-1}\frac{2}{3}\right)$  ( in exact form ) **2**

c) Differentiate the following:

(i)  $\sin^{-1}(2x + 1)$  **1**

(ii)  $x^2 \log_e(\tan^{-1} x)$  **2**

d) For the function given below:



(i) State the domain for which  $f(x)$  will have an inverse function. **1**

(ii) Sketch  $y = f^{-1}(x)$ , clearly showing the domain and range. **2**

**End of Question 2**

**Question 3** (14 marks) [BEGIN A NEW PAGE]

a) Find the indefinite integral  $\int \frac{dx}{\sqrt{3-x^2}}$  **1**

b) Evaluate  $\int_0^{\frac{2}{3}} \frac{1}{4+9x^2} dx$  **2**

c) It is given that if  $f(x) = x \tan^{-1}(x) - \frac{1}{2} \log_e(x^2 + 1)$ , then  $f'(x) = \tan^{-1}(x)$

Use this information to find the area enclosed by  $y = \tan^{-1}(x)$ , the  $x$ -axis and the ordinates  $x = -1$  and  $x = 1$  (Give your answer in simplest exact form) **3**

d) Using the substitution  $u = e^x$ , show that

$$\int_{-1}^1 \frac{e^x}{1+e^{2x}} dx = \tan^{-1}\left(\frac{e^2-1}{2e}\right).$$
 **3**

e) (i) State the domain and range of the function **2**

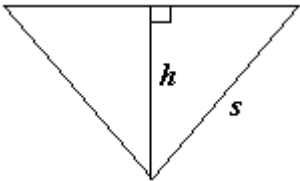
$$f(x) = \tan^{-1}(\sqrt{x}).$$

(i) Find  $f'(x)$  and comment on the behaviour of  $f'(x)$ , for  $x \geq 0$ . **2**

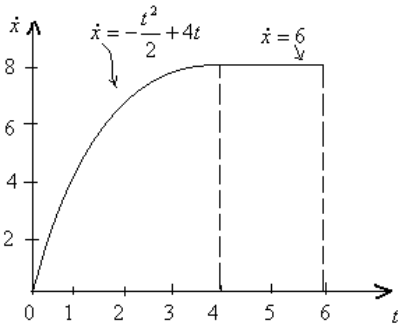
(iii) Hence, draw a sketch of  $y = f(x)$ . **1**

**End of paper**

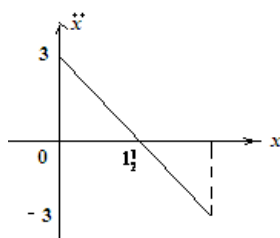
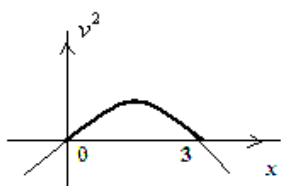
Year 12 Extension 1 Task 3: Marking Criteria

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                              |
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| <p>a)(i) <math>v = -2\sqrt{1-x^2}</math></p> <p><math>t = 0, x = 0</math></p> $\frac{dx}{dt} = -2\sqrt{1-x^2}$ $\therefore \frac{dx}{\sqrt{1-x^2}} = -2dt$ $\sin^{-1}x = -2t + C$ $t = 0, x = 0 \Rightarrow C = 0$ $\therefore \sin^{-1}x = -2t$ $\therefore x = \sin(-2t)$ $x = -\sin(2t)$ <p>(ii) when <math>t = \frac{\pi}{6}, x = -\frac{\sqrt{3}}{2}</math></p> $v = -2\sqrt{1-x^2}$ $= -2\sqrt{1-\frac{3}{4}}$ $= -1 \text{ ms}^{-1}.$ | <p><b>1 mark for expressing <math>v</math> as <math>\frac{dx}{dt}</math> and getting the correct integral <math>\sin^{-1}x = -2t + C</math></b></p> <p><b>1 mark for evaluating <math>C</math> as getting the correct expression for <math>x</math>.</b></p> <p><b>1 mark for substituting into the expression for <math>x</math> in (i) and evaluating <math>x</math>.</b></p> <p><b>1 mark for correct substitution to <math>v</math> and</b></p> | <p><i>Many students forgot to make <math>x</math> subject of the formula.</i></p>            |
| <p>(b)</p>  <p>(i) <math>S = \frac{h}{\sin 60^\circ} = \frac{2h}{\sqrt{3}}</math></p> <p>Volume of the trough =</p> $\frac{1}{2} \times sh \times 10$ $= \frac{1}{2} \times \frac{2h}{\sqrt{3}} \times h \times 10$                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p><i>“Show that” question must have every line of verbal and numerical working out.</i></p> |



|                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                            |
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|  <p>(ii) Distance travelled = Area under the velocity curve.</p> $D = A_1 + A_2$ $= \int_0^4 -\frac{t^2}{2} + 4t dt + 2 \times 8$ $= \left[ -\frac{t^3}{6} + 2t^2 \right]_0^4 + 16$ $= 21\frac{1}{3} + 16 = 37\frac{1}{3} \text{ metres}$                           | <p><b>1 mark: For the graph</b></p> <p><b>1 mark: Evaluates area <math>A_1</math> (area under the parabola).</b></p> <p><b>1 mark: Evaluates area <math>A_2 = 16</math> sq. units and adds it to <math>A_1</math>.</b></p>                                                                                                                             | <p><i>Many students did not get this graph correct. Even those who got it, did not have sufficient information on it. You absolutely needed to have y-intercept 8. Y-intercept of 4 was very common. Students thought that the y-intercept of <math>\dot{x}</math> and <math>\ddot{x}</math> are exactly the same.</i></p> |
| <p>d) (i) <math>\ddot{x} = 3 - 2x</math>.</p> $\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 3 - 2x.$ <p>Integrating both sides w.r.to <math>x</math>,</p> $\frac{1}{2} v^2 = 3x - x^2 + C$ <p><math>x = 1, v = 2 \therefore C = 0</math></p> $v^2 = 6x - 2x^2$ $v^2 = 2x(3 - x)$ <p><math>x = 1, v = 2</math></p> $\therefore v = \sqrt{2x(3 - x)}$ | <p><b>1 mark: sets <math>\ddot{x} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)</math> and integrates w.r.to <math>x</math> to get</b></p> $\frac{1}{2} v^2 = 3x - x^2 + C$ <p><b>1 mark: Evaluates <math>C</math> and gets the expression <math>v = \sqrt{2x(3 - x)}</math></b></p> <p><i>Must explain why only the positive root was chosen.</i></p> |                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                            |

(ii)



$$v^2 \geq 0 \quad \therefore 0 \leq x \leq 3.$$

$$\ddot{x} = 3 - 2x$$

$$\text{And } v = \sqrt{2x(3-x)}$$

- At  $x = 1$ ,  $v = 2$  means that the particle will move right speeding up ( $v > 0$  and  $\ddot{x} > 0$ ) attains maximum speed at  $x = 1 \frac{1}{2}$  ( $\ddot{x} = 0$ ).
- For  $x > 1 \frac{1}{2}$ ,  $v > 0$  and  $\ddot{x} < 0$ ,  $\therefore$  the particle slows down until it comes to rest at  $x = 3$ . But the force ( $\ddot{x}$ ) acting on it is negative,  $\therefore$  the particle moves to the origin. I.e. The particle oscillates between 0 and 3.
- $\ddot{x} = 3 - 2x$   
 $= -2 \left( x - \frac{3}{2} \right)$  is of the form  $-n^2 (x - b)$ .  
 $\therefore$  the motion is SHM.

*Very poorly done.*

*When describing motion, you must comment as  $x$  increases, how are  $\dot{x}$  and  $\ddot{x}$  are affecting the motion.*

*Follow the particle (make the comments progressively) to describe motion; not picking points here and there and giving fragmented description.*

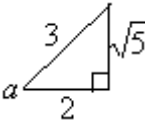
**1 mark for point**

**1 mark for point 2**

**1 mark for point 3**

## Question 2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
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| <p>(a) (i) <math>y = 25 + 18 \cos \left[ \frac{\pi}{10}(t+9) \right]</math></p> $\dot{y} = -18 \sin \left[ \frac{\pi}{10}(t+9) \right] \times \frac{\pi}{10}$ $\ddot{y} = -18 \left( \frac{\pi}{10} \right)^2 \cos \left[ \frac{\pi}{10}(t+9) \right]$ $= -18 \left( \frac{\pi}{10} \right)^2 \left( \frac{y-25}{18} \right)$ $= - \left( \frac{\pi}{10} \right)^2 (y-25)$ <p>This is SHM about <math>y = 25</math>.</p> <p>(ii) <math>-1 \leq \cos \left[ \frac{\pi}{10}(t+9) \right] \leq 1</math></p> $\therefore y_{\max} = 25 + 18 = 43$ $y_{\min} = 25 - 18 = 7$ $\therefore D = 36$ $B = 7$ <p>(iii) Period = <math>\frac{2\pi}{\frac{\pi}{10}} = 20</math> sec</p> $\therefore \text{No: of revolutions/min} = \frac{60}{20}$ $= 3$ | <p><b>1 mark: for correct velocity expression</b></p> <p><b>1 mark: manipulating to get <math>\ddot{y} = -n^2(x-b)</math> and hence SHM.</b></p> <p><b>1 mark: for correct D</b></p> <p><b>1 mark: for correct B</b></p> <p><b>(Reduce 1 mark if <math>-1 \leq \cos \left[ \frac{\pi}{10}(t+9) \right] \leq 1</math> is not given)</b></p> <p><b>1 mark: for correct period</b></p> <p><b>1 mark: correct no: of revolutions using the period from (ii)</b></p> |  |
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|                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                 |  |
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| <p>b) (i) <math>\frac{\pi}{3}</math></p> <p>(ii) Let <math>a = \cos^{-1} \frac{2}{3}</math></p> <p>then, <math>\cos a = \frac{2}{3}</math></p> <p><math>\tan a = \frac{\sqrt{5}}{2}</math></p> <p><math>\therefore \tan(\cos^{-1} \frac{2}{3}) = \frac{\sqrt{5}}{2}</math></p>  | <p><b>1 mark for correct answer</b></p> <p><b>1 mark for the value of <math>\cos a</math> and the representation with a diagram</b></p> <p><b>1 mark for correct <math>\tan a</math> using the diagram.</b></p> |  |
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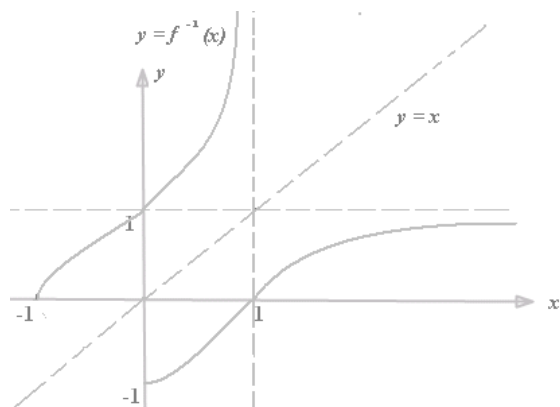
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| <p>c)(i) <math>\frac{d}{dx} [\sin^{-1}(2x+1)]</math></p> <p><math>\frac{1 \times 2}{\sqrt{1 - (2x+1)^2}}</math></p> <p>(ii) <math>\frac{d}{dx} [x^2 \log_e (\tan^{-1} x)]</math></p> <p><math>u = x^2 \quad u' = 2x</math></p> <p><math>v = \log_e (\tan^{-1} x) \quad v' = \frac{1}{\tan^{-1} x (1+x^2)}</math></p> <p><math>\therefore \frac{d}{dx} [x^2 \log_e (\tan^{-1} x)] =</math></p> <p><math>\frac{2x \times [\log_e (\tan^{-1} x)] + x^2}{\tan^{-1} x (1+x^2)}</math></p> | <p><b>1 mark</b></p> <p><b>1 mark for correct differentiation of u and v.</b></p> <p><b>1 mark for applying product rule correctly.</b></p> |  |
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d) (i) Largest possible domain

$$D: x \geq 0$$

**1 mark**

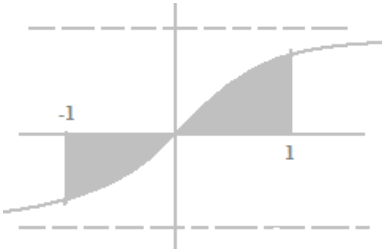
(ii)



**1 mark:** *correct domain and range of  $y = f^{-1}(x)$*

**1 mark:** *correct shape of  $y = f^{-1}(x)$  including the correct asymptote*

### Question 3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) <math>\int \frac{dx}{\sqrt{3-x^2}} =</math><br/> <math>\sin^{-1}\left(\frac{x}{\sqrt{3}}\right) + C</math></p>                                                                                                                                                                                                                                                                                                                                                                      | <p><b>1 mark for correct answer</b></p>                                                                                                                                                                   | <p><i>Well done except that some students forgot to <math>\sqrt{3}</math> and TSI</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>b) <math>\int_0^{\frac{2}{3}} \frac{1}{4+9x^2} dx</math><br/> <math>= \int_0^{\frac{2}{3}} \frac{1}{9\left(\frac{4}{9}+x^2\right)} dx</math><br/> <math>= \frac{1}{9} \times \frac{3}{2} \left[ \tan^{-1}\left(\frac{3}{2}x\right) \right]_0^{\frac{2}{3}}</math><br/> <math>= \frac{1}{6} \{ \tan^{-1}(1) - \tan^{-1}(0) \}</math><br/> <math>= \frac{1}{6} \times \frac{\pi}{4} = \frac{\pi}{24}</math></p>                                                                          | <p><b>1 mark for correct integration</b></p> <p><b>1 mark for correct evaluation</b></p>                                                                                                                  | <p><i>Reasonably well done. The formula is on the table of standard integrals.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p>c)</p>  <p>Required area <math>= \int_{-1}^1 y dx</math><br/> <math>= \int_{-1}^1 \tan^{-1}(x) dx</math><br/> <math>= 2 \times \int_0^1 \tan^{-1}(x) dx</math><br/> <math>= \frac{1}{2} [x \tan^{-1}(x) - \log_e(x^2 + 1)]_0^1</math><br/> <math>= \frac{1}{2} \left[ \left( \frac{\pi}{4} - \ln(2) \right) - (0 - \ln 1) \right]</math><br/> <math>= \frac{\pi}{8} - \frac{1}{2} \ln(2)</math></p> | <p><b>1 mark for understanding it is an odd function and therefore multiplying by 2</b></p> <p><b>1 mark for using the correct integral from the data</b></p> <p><b>1 mark for correct evaluation</b></p> | <p><i>Students who took the time to draw a diagram did this well. They quickly realised that they could double the integral from 0 to 1. Students who tried to take the absolute value of the interval from -1 to 0 got into a mess.</i></p> <p><i>Students who struggled to attempt this question are reminded that a quick diagram can prompt thought processes</i></p> <p><i>Students who chose to ignore the data and take the area between the curve and the y axis were unsuccessful and got into a mess.</i></p> <p><b>USE THE INFORMATION</b></p> |

d)  $\int_{-1}^1 \frac{e^x}{1+e^{2x}} dx$

Let  $u = e^x$

$$\therefore du = e^x dx$$

$$x = -1, \quad u = \frac{1}{e}$$

$$x=1, \quad u=e$$

$$\int_{\frac{1}{e}}^e \frac{du}{1+u^2}$$

$$= \left[ \tan^{-1}(u) \right]_{\frac{1}{e}}^e$$

$$= \tan^{-1}(e) - \tan^{-1}\left(\frac{1}{e}\right)$$

Let  $k = \tan^{-1}(e) - \tan^{-1}\left(\frac{1}{e}\right)$

$$\therefore \tan k =$$

$$\tan\left(\tan^{-1}(e) - \tan^{-1}\left(\frac{1}{e}\right)\right)$$

$$= \frac{e^{-\frac{1}{e}}}{1 + e \times \frac{1}{e}}$$

$$= \left( \frac{e^2 - 1}{2e} \right)$$

$$\therefore k = \tan^{-1} \left( \frac{e^2 - 1}{2e} \right).$$

**1 mark for correct conversions**

**1 mark for correct integration**

**1 mark for the correct solution**

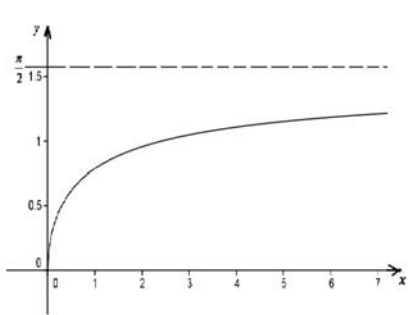
*Students who changed the terminals in the start of the question were more successful.*

*The first part of the question was well done.*

*Students are reminded that SHOW means you need to show every step of working.*

*You cannot factorise inverse trig functions nor use a mutation of log laws for inverse trig*

*The third part of this question was not done well.*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>e)</p> $f(x) = \tan^{-1}(\sqrt{x})$ <p>(i) Domain: <math>x \geq 0</math> <b>1 mark</b><br/> Range: <math>0 \leq y &lt; \frac{\pi}{2}</math> <b>1 mark</b></p> <p>(ii) <math>f'(x) = \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}}</math><br/> <math>= \frac{1}{2\sqrt{x}(1+x)}, x \neq 0</math> <b>1 mark</b></p> <p>Since, <math>x &gt; 0</math>, <math>f'(x) &gt; 0</math>,<br/> hence, <math>f(x)</math> is increasing<br/> from <math>x = 0</math> to <math>\frac{\pi}{2}</math>, <math>y \neq \frac{\pi}{2}</math>. <b>1 mark</b></p> <p>(iii) <math>f'(x)</math> undefined at <math>x = 0</math> and<br/> <math>f'(x) \rightarrow \infty</math>, as <math>x \rightarrow \infty</math>.</p>  | <p>(i)<br/> <b>1 mark: for correct domain</b><br/> <b>1 mark: for correct range</b></p> <p>(ii)<br/> <b>1 mark for <math>f'(x)</math></b></p> <p><b>1 mark stating <math>f'(x) &gt; 0</math> in the domain given and determines corresponding <math>f(x)</math> values. I gave 1 mark for <math>f'(x) &gt; 0</math> and <math>x \neq 0</math></b></p> <p>(iii)<br/> <b>Sketch 1 mark.</b><br/> <b>Must show vertical gradient at <math>x = 0</math> and asymptote.</b></p> | <p>(i) Students are reminded to always notice square roots and denominators!! Many students could find the domain but failed to see the effect of a change in domain on the range. Many students forgot that there is an asymptote on <math>\pi/2</math> so wrote the domain as <math>\leq \pi</math></p> <p>(ii) many students could differentiate this correctly and obtained 1 mark, but some forgot to differentiate the <math>\sqrt{x}</math>. The second part of this was very poorly done. Many students commented on <math>f(x)</math>. Again, square roots and denominators!!</p> <p>(iii) One student obtained the 1 mark for this Q. Students seem to have no idea what an undefined gradient means. We need to get them to find the gradient of say, <math>y=4</math>. Many students were able to obtain the correct shape of the graph and the asymptote.</p> |
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